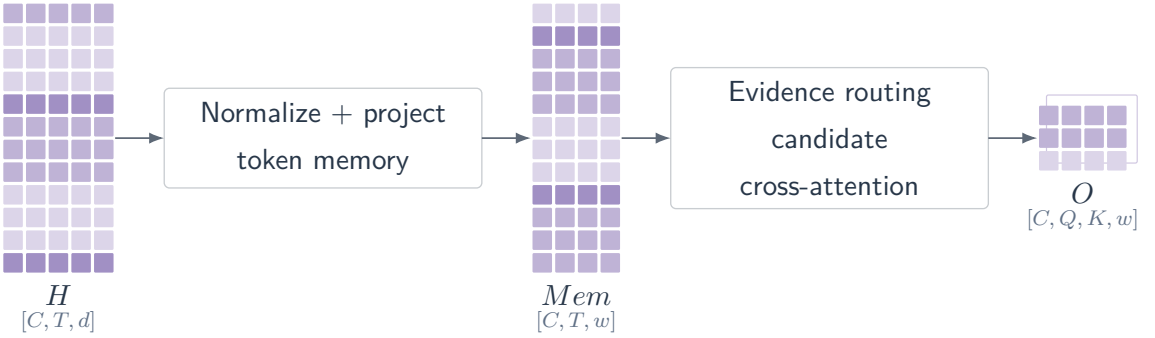
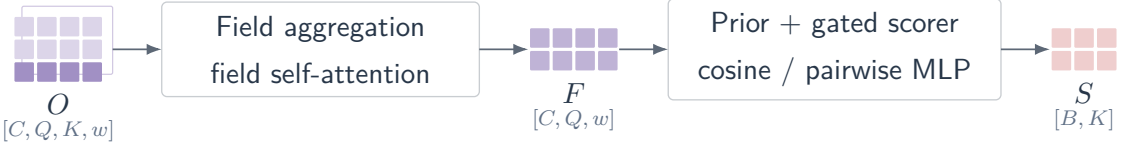
$$O = \mathrm{Route}(O^0; Mem), \quad F = \mathrm{DecoderHead}(F^0; Mem), \quad S_{f,k} = \pi_{f,k} + \sigma(\gamma)\, g(F_f, O_k)$$

Build token memory and route evidence into candidate representations



Aggregate each field, allow field interaction, then score field–candidate pairs



Axes $H : [C, T, d]$; $Mem : [C, T, w]$; $O : [C, Q, K, w]$; $F : [C, Q, w]$; $B = CQ$ in the fixed-size example. Leading C, Q axes flatten before reporting $S : [B, K]$.

Meaning This comparison family adds structured interactions in the head. Candidates may affect each other through field aggregation even without candidate self-attention. Causal visibility in the backbone and visibility in the head must be analyzed separately.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.